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SPECIAL ISSUE: ADVANCING SOCIO-HYDROLOGY

Socio-hydrology with hydrosocial theory: two sides of the same coin?

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ABSTRACT
This paper reviews socio-hydrology and hydrosocial research, finding a sophisticated relationship with emergent syntheses. We examined 419 papers by topic, region of study, theories implemented, journal, and year published to ascertain trends in both subfields. We found important overlap and considerable difference between subfields. Whereas hydrosocial research took years to develop, socio-hydrology commenced with an inaugural paper in 2012. While the former focuses on power and scale in studying water demand, the latter concentrates on practical responses to climate extremes. Hydrosocial research usually relies on qualitative methods, and socio-hydrology research the quantitative. In the geographic regions where the former does not focus, the latter does. The former often relies on post-structuralist theory, whereas the latter uses positivist approaches. Our review concludes that socio-hydrology and hydrosocial research exist in a complex epistemological relationship, offering fertile grounds for lively discussions from which both will continue to benefit.

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Introduction

Engaging with the literature on coupled human–water systems aids in analyses of water availability, quality, hazards, and related ecosystem services. As anthropogenic climate change places a strain on water resources around the world, the way scholars, policymakers, and resource managers understand such systems will prove instrumental to ensuring the sustainability of threatened communities and defusing potential social conflict (Cisneros *et al.* 2014). Without comprehending those interactive relationships, appropriate human responses to challenges and hazards involved in hydrologic systems become effectively impossible. Yet, the literature conceptualizing human and water interactions appear divided at times between those adopting a "hydrosocial" approach that centers power and scale, on the one hand, and those using "socio-hydrology" to effectively ascertain and adapt to specific needs from a positivist perspective on the other.

There has been one key literature review proposing collaboration between the two subfields by Wesselinck *et al.* (2017) and another review proposing "*interdisciplinary water resource geography*" as a way of bringing socio-hydrology and critical geographies of water under the same umbrella (Rusca and Di Baldassarre 2019). The virtues of the Wesselinck *et al.* (2017) paper are myriad, including discussions about the crucial foundations of both subfields, distinguishing the two before calling for collaboration. In particular, their review extensively describes background information on Earth System Sciences as a kind of formative basis for socio-hydrology. Wesselinck *et al.* (2017) also offer the valuable conceptual heuristic of "narrative" as a binding heuristic to promote collaboration across the divide, through which both disciplines can develop a (post) positivist grounding.

At the same time, Rusca and Di Baldassarre (2019) call for the integration of socio-hydrology with critical geographies of water (which encompasses hydrosocial theory) by accounting both for their own "*moral obligation as scientists aspiring to change (rather than interpret) the world*" and "*the mutual shaping of society and hydrological flows*" (p. 10). While the Wesselink review was published in 2017 when the number of socio-hydrological papers numbered 69 articles, today that number has more than tripled, giving rise to new innovations and epistemologies described by Rusca and Di Baldassarre (2019). Furthermore, by situating hydrosocial studies' discrete epistemological origins in critical geographies of water as long ago as the late-1990s, Rusca and Di Baldassarre (2019) identify different theoretical commonalities with socio-hydrology. Lastly, Rusca and Di Baldassarre (2019) importantly caution that "[p]*erceptions of the irreconcilable differences and of asymmetrical collaborations between natural and social sciences deter opportunities of meaningful collaborations*" (p. 2).

We offer a further analysis of integration in keeping with both these review papers, insofar as the careful formulation of narratives building on practical case studies and "*preserving methodological and epistemological differences*" can synthesize the subfields (Rusca and Di Baldassarre 2019, p. 9). While both subfields study different aspects of the coevolution human–water systems, in doing so, they have developed strengths that compensate for what the other may lack. At the same time, our study finds encouraging collaboration occurring amid the dynamic relationship between the discrete subfields, indicating shared strengths and mutual aid. Without elucidation of such developments, alienation between interdisciplinary fields of research can lead to marginalization of important ideas, despite the increasing interest in the subject of coupled

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human–water research resulting from growing challenges of climate and urbanization.

The present review manifests an effort to comprehensively analyze both subfields of research, taking stock of their differences, and pointing to contributions from both sides that ultimately describe crucial intersections. We begin by examining the basis for socio-hydrology, interrogating its claim to the mantle of a "new science" coupling human–hydrologic systems. We situate its origin in the relatively recent site of coupled human–hydrologic studies in the late 1970s, discerning four discrete tendencies that have since emerged – hydrosociology, critical water studies, hydrosocial theory, and socio-hydrology. We then discuss critical water studies and its influence on fundamental premises of hydrosocial theory, followed by a review of the available hydrosocial literature. After investigating hydrosocial literature, we review the available literature on socio-hydrology, carefully examining its complex relationship with hydrosocial studies. Finally, we discuss, in depth, the contours of hydrosocial studies and socio-hydrology together, finding both important distinctions and significant intersections. By describing the complexity of the continued relationship between subfields with empirical evidence, we hope to widen the available tools and approaches possible within a general theoretical position involving the co-evolution of human–water systems. It is necessary to understand these subfields together if we are to use the strengths of both to achieve water sustainability.

Methods

We used a mixed methods approach to discern not only topics but also the methods used and study sites researched. To start, we downloaded the citations and abstracts for every article produced by a topic search in Web of Science for "hydrosocial" or "hydro-social" (n = 207) on 20 January 2020 and analyzed them for thematic content, study site, and theoretical framework. We performed the same analysis on the 212 articles yielded from a Web of Science topic search of "socio-hydrology" during the same time.

Firstly, we tracked the increase in interest in these subfields over time by creating a timeline of the number of articles published with the topics "hydrosocial," "hydro-social," or "socio-hydrology." We determined the main methods and theories utilized by each subfield and compared them to gain a sense of the overlap occurring over time. This process required a qualitative study of the journal abstracts and, in many cases, the full articles, to facilitate a more robust exposition of the central themes and ideas of each subfield. We also developed graphs to describe the topical focus of each subfield over time, based on Water Demand, Quality, Climate Extremes, and Ecosystems Services. While there are some overlapping papers in these four topics, we chose the prevailing idea through content analysis. Viewing the changing topical courses of these subfields and their relative methods and theories over time helped to discern inflection points and overlap.

Lastly, we counted the numbers of studies per country and created a choropleth map showing the most-frequent study sites for each subfield. We then used data based on study topic, categorized according to the aforementioned four main themes, to create pie charts. Those pie charts were fitted as graduated symbols into the choropleth map to indicate both number and type of studies. We believe that this methodology breaks new ground in the field of coupled human–water systems by providing the most comprehensive quantitative and spatial analysis of the state of the research to date, as well as a useful qualitative review of their major theoretical positions.

Socio-hydrology, hydrosociology, or hydrosocial theory?

Locating the origins of hydrosocial theory and its precursor, critical geographies of water, may prove impossible, but the work of Wittfogel often serves as a useful starting point in the modern era (Wittfogel 1957, Banister 2014, Linton and Budds 2014). Although not embraced by everyone, Wittfogel's assessment of the connection between political organization and hydrology provided the basis for adroit analyses of coupled human–water systems in the American West (Meisner 1963, Worster 1985). Perhaps even more pertinent to this paper, a term approximate to "hydrosocial" or "socio-hydrology" emerged in the work of Falkenmark, who opened her path-breaking 1979 article, "Main Problems of Water Use and Transfer of Technology," with the sentence, "*Man and water are closely related to each other in a dualistic manner*" (p. 435).

For Falkenmark, water serves as life-sustaining and producing while providing quality of life and symbolic value. The focus of Falkenmark's "hydrosociology" becomes the forecasting of potential issues with regards to human–water systems. As well, Falkenmark posits that a transfer of technologies will break through regional limitations and develop shared capacity for overcoming crisis. While others had written in the same vein as "hydrosociology" on the co-evolution of society, political organization, and hydrological systems (e.g. Wittfogel), Falkenmark's incisive commentary joined a sense of purpose to scientific methodology, laying the groundwork for today's efforts. Such efforts joined the development of socio-ecology in the 1980s and 1990s to promote studies of ecosystems that included social influences, adaptive capacity, and resilience. The more recent advent of a hydrosocial and a socio-hydrological field of research, however, remains both within but also outside of this course of study (Pande and Sivapalan 2017).

Hydrosocial research as practiced today comes largely from the junction of Political Ecology and Science and Technology Studies. Formed in the late-1990s through the work of a number of theorists scattered across the world, the concept of the hydrosocial cycle emerged slowly with vigorous discussions about hybridity and technology (Fig. 1). We locate the earliest threads in Bakker's work on flooding in England (2000) and Turton and Meissner (2002) on the "*hydrosocial contract*," along with Swyngedouw's exploration of "*how the circulation of water is embedded in the political ecology of power, through which the urbanization process unfolds*" (1997, p. 313). Swyngedouw writes of "*nature's water*" and a "*historical geography of water control*" through which water becomes domesticated. Although Swyngedouw did not name "hydrosocial" relations in 1997, as Bakker (2000), Warner

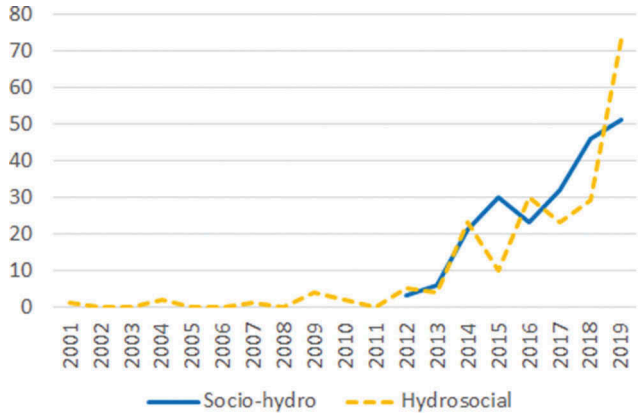

Figure 1. Trend in hydrosocial studies and socio-hydrology, 2001–2019.

(2000), and Turton and Meissner (2002) did, the former created the fundamental premises on which hydrosocial studies would develop by conceptualizing the complexity of power relations relative to water in terms of the economic, social, spatial, and political aspects of ecology.

On the other hand, socio-hydrology emerged first in 2012 with the near-simultaneous publication of two unrelated articles: "Irrigation and Development in the Upper Indus Basin" in *Mountain Research and Development* and "Socio-hydrology: A new science of people and water" in *Hydrological Processes*. The former paper, written by Nusser *et al.* (2012), offered a "*socio-hydrological framework* [that] *encompasses all dimensions and factors that are crucial for an integrated analysis of irrigated land use patterns and corresponding land cover changes*" (p. 60). Sivapalan *et al.* (2012) published their paper just two months later and aimed more generally at "*understanding the dynamics and co-evolution of coupled human–water systems*" (p. 1271). Both adopted a similar outlook, but Sivapalan *et al.* (2012) used a general approach that extended beyond just irrigation and land cover. However, like hydrosocial studies, which emerged through urban political ecology, socio-hydrology has traditionally focused on urban or rural areas rather than exurban or periurban areas (there are exceptions, e.g. Peloso and Harris 2017, Roth *et al.* 2019).

The two subfields have published different topics and publication outlets. Water resources dominates the research areas feeding into socio-hydrology, incorporated into 54% of articles gained through our topic searches. Meanwhile, "hydro-social" or "hydrosocial" studies show greater distribution between the topics "geography" (34%) and "water resources" (36%). Most articles (11.1%) of the hydrosocial persuasion have been published in *Geoforum*, with 10.6% coming through *Water International*. This is likely due to a special issue of *Geoforum* dedicated to hydrosocial theory in 2014 and two in *Water International* in 2016 and 2019. Another special issue of *Water* was devoted to hydrosocial theory in 2019. On the other hand, 18.8% of socio-hydrological research comes by way of *Water Resources Research*, while *Hydrology and Earth System Science* published 13.6%, followed by *Hydrological Sciences Journal* (8.5%) and *Journal of Hydrology* (7%). Research areas were again dominated by Water Resources (70.4%), but Environmental Sciences took second place (41.3%) with Multidisciplinary Geosciences (25%) and Limnology (18.8%) coming next.

Through this analysis, it would appear that both subfields deal with Water Resources, but each in their own way and from nominally different research communities represented by different scholarly publications. Hydrosociology and socio-ecology might be seen as important taproots, but the epistemologies of hydrosocial research and socio-hydrology are distinct, as illustrated by the different composition of publication networks. We can safely conclude, then, that the two subfields are unique but intersecting and must first be examined on their own terms.

Origins of hydrosocial studies

Early hydrosocial research developed through critical geographies of water, conceiving of new ways to understand the coevolution of human and water systems. The "*hydrosocial cycle*" (Bakker 2000, 2003) ideated an inextricable relationship between the hydrological cycle and human societies, while the "*hydrosocial contract*" connoted the means through which technonatures deterritorialize water and integrate it within socio-political regimes of sanitation and water access underwritten by the government's responsibility to provide water to its citizens and protect them from water hazards like flooding (Warner 2000, Turton and Meissner 2002, Cantor 2017). These formative studies of the hydrosocial cycle carved out a new niche within critical geographies of water, which further developed through empirical work mobilizing the term into a subfield of research.

Hydrosocial theory emerged from a heterodox assemblage of Marxian understandings of capitalist accumulation and development in correspondence with multifarious theories of power and scale often described as "*post-structuralist*" (e.g. Haraway 1991). According to this approach, the accumulation not only of capital but of power manifests spatial conditions for "*uneven development*" (Smith 1984) as a result of a "*spatio-temporal fix*" (Harvey 2006), whereby the overaccumulation of capital in the "interior" requires the transformation of the ecological "exterior" into productive sites of primary extraction (e.g. mines, timber, oil wells) to maintain an unsustainable "*metabolic rate*" of urban-industrial development (Foster 2000). Here, landscape and cultural geography enters into critical relation with geographies of development and resource management through "*waterscapes*," hybrid constructs that are "*part natural and part social*," embodying "*a multiplicity of historical-geographical relations and processes*" of internalization and externalization (Swyngedouw 1999, p. 445, see also Molle *et al.* 2009, p. 2).

In relation to capital, these co-evolutionary relations and processes reify systems of inequality while manufacturing out of "*nature*" a "*second nature*" through natural resource management (Cronon 1991). As a result, the urban centers of capital determine the phenomenological value of the "*world*," vis-a-vis what Marx calls the "*social process of production*" (Marx 1999). This world-making process extends through networks, understood in the post-modern sense as natural systems semiotically mediated by way of representational

networks that produce denatured social assemblages often determined toward a reconstructed socio-ecological hybridity (Latour 2004). Hence, in Maria Kaika's words, "*the 'world' is a historical-geographical process of perpetual metabolism in which 'social' and 'natural' processes combine in a historical-geographical 'production process of socio-nature' whose outcome (historical nature) embodies chemical, physical, social, economic, political and cultural processes in highly contradictory but inseparable manners*" (2005, p. 23).

Hydrosocial theory often views unequal water access and sanitation as products of two distinct regimes of power that determine the governance of health and wellbeing for the rich and poor through Foucault's ideation of biopolitical divisions between "*haves and have nots*" (Cantor 2017). According to Foucault, by mobilizing political economy to determine the fundamental socio-political arrangements of the population, the state can "*conduct the conduct*" of the population through dispersed manifestations of power without relying on a central authority that can be easily located and resisted (2008). Thus, hydrosocial theorists argue, the urban hydrosocial cycle manifests a process of dispossession and disenfranchisement incumbent on the spatial transformation of water from "*nature's water*" to water that can hold a "*duty*" toward humanity and that can become "*wastewater*" (Swyngedouw 2004). That world-making metabolic process through which humanity engages with waterscapes and is, in turn, changed by its efforts Swyngedouw names "*coevolution*" (2006).

Focusing on feedback systems between human and water interactions, the hydrosocial cycle identifies the human influence on the hydrologic cycle as part of a dialectical development of waterscapes and social systems (Swyngedouw *et al.* 2002). Linton (2008) argues for a new way of understanding the study of water outside the scientifically constructed and abstract parameters of hydrology in order to gain perspective on its connection to "*the old, supply-oriented paradigm of water resource management*" (p. 642). It follows that redefining the hydrologic cycle to include the challenges of the Anthropocene would entail "*the integration of physical, biological, biogeochemical, and human components of a more general 'global water system'*" (p. 645). Published the next year, Budds (2009) identified the limitations of hydrological assessments, drawing on critical geographies of water to describe the hydrosocial cycle as a means of extending the production of knowledge beyond technical experts and "*exploring the production and use of hydrological data*" (2009, p. 420). Linton and Budds (2014) refined these approaches into a "*relational-dialectical*" model, challenging humanity's sovereignty over water by highlighting the *internal* relation between water and social power, thus identifying human–water systems as hybrids of nature and society.

That year, Boelens introduced the notion of "*water truths*," ways that societies organize around water and change water by ascribing "*different cosmological pathways*" that "*form a socionatural network traveled by gods and ancestors, engendering the human world*" (2014, p. 243). Neoliberal practices deterritorialize the hydrosocial cycle and strip it of the truths that cohere to social practices, altering its socio-natural metabolism and extending it into transnational networks mediated through the global market (Boelens 2014, Bakker 2000, Swyngedouw and Heynen 2003). Under this neoliberal order, hydrosocial theory posits, the "*post-political*" rescaling of governmental authority upward to multinationals and downward to consensus-based non-governmental organizations accompanies privatization and deregulation (Bakker 2003, Budds and McGranahan 2003, Castro 2007). Against this tendency, Swyngedouw calls for "*a rethinking of the meaning of citizenship to recognize the multiplicity of identities, the rhizomatic meanderings of meanings, practices, and lives*" (2007, p. 23). Hence, hydrosocial research demands a transition from technocratic urban policy into more direct, democratic processes involving debate and contention across "*multiscalar networks*." (Boelens 2008).

Although Swyngedouw, Boelens, and Kaika provided some of the most important initial pushes, we find Bakker's early and continued influence from critical geographies of water noteworthy. Boelens appears to be the most published author in hydrosocial studies with 23 articles on the topic. Other key authors in the subfield include Linton, Budds, and Hoogesteger. The prominent position Boelens takes in the literature also helps to determine the most prolific study site – his focus areas become the most studied, particularly Peru with 17 published studies. Swyngedouw's key study area of Spain comes in second place with 16 published studies. In general, South America is the most studied with 39 in total, most of which come from the Andean region. By comparison, there are 11 hydrosocial studies from Africa and 22 from Asia. As many as 59 articles did not clearly attribute their study to a particular site, many of which preferring to make broader assessments on more theoretical grounds.

Reviewing hydrosocial theory

Out of 207 records for hydrosocial theory, according to our assessment, the most frequent foci discernable from abstracts included irrigation (35 articles), scarcity (24), dams (23), groundwater (11), desalination (10), glaciers (7), sanitation (6), and mines (5). While 10 articles focused on history, hydrosocial theory as a subfield of coupled human–water research concentrates mostly on contemporary water governance issues (Fig. 2). Roughly one-third of all hydrosocial texts incorporate governance as a key topic. Nineteen of those articles were theoretical interventions, while the rest involved place-based analyses from Canada (Cook 2016, Stevenson 2018) to Spain (Swyngedouw 2007, Sanchis-Ibor *et al.* 2017, Duarte-Abadia *et al.* 2019) to Peru (Carey *et al.* 2014, Yacoub *et al.* 2016, Mark *et al.* 2017, Damonte and Boelens 2019).

The theoretical frameworks of hydrosocial studies have drawn extensively from the work of Swyngedouw, Boelens, and Kaika described above, incorporating post-structuralist theory (Hoogendam and Boelens 2019, Valladares and Boelens 2019) with analyses of coupled socio-ecological systems (Lerner *et al.* 2018, Carey *et al.* 2014). This context has provided intriguing opportunities for inquiries into "*multiscalar networks*" (Hommes *et al.* 2016, Boelens *et al.* 2019) and decolonization (McLean 2017, Stevenson 2018, Cavazos Cohn *et al.* 2019, Duarte-Abida and Boelens 2019). An entire book informed by hydrosocial studies but not included in the Web of Science dragnet, *Negotiating Water Governance: Why*

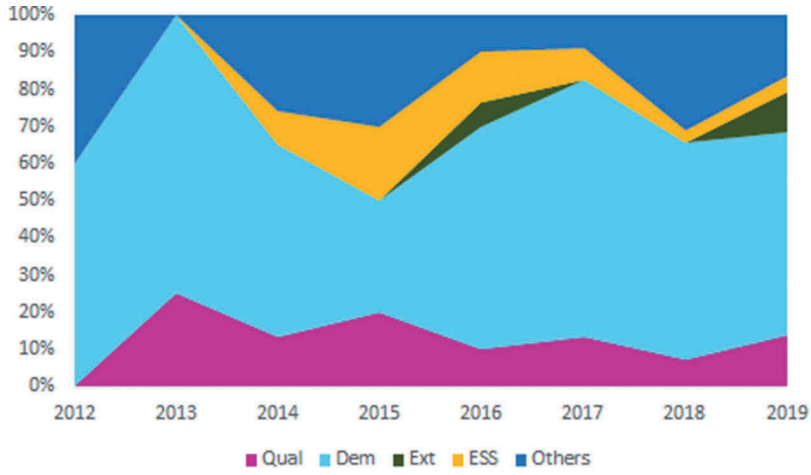

Figure 2. Percent distribution of hydrosocial studies by topic: Extremes (purple), Demand/Use (blue), Quality (orange), and Ecosystems Services (green).

the Politics of Scale Matters, assesses rescaling political power in watersheds, often through the efforts of indigenous groups (Norman *et al.* 2015).

The usage of scale in hydrosocial theory generally returns to the specific context of multi-scalar networks exposited by Boelens in 2008. These networks can be conceived simply as complex and pluralistic assemblages of social organizations, often including disenfranchised groups organizing in opposition to the state and corporations (Hoogesteger and Verzijl 2015, Stoltenborg and Boelens 2016, Truelove 2019). However, the hydrosocial literature on multi-scalar networks also involves power relations within and between administrative bodies, legal arrangements, and physical structures (Hommes and Boelens 2017). For this reason, multi-scalar networks have been used to describe complex systems with a broad range of variables acting in relation to dynamic power relations that assemble nested hierarchies depending on spatio-temporal conditions. Hommes and Boelens describe multi-scalar networks as comprising "*legal-political and social institutions, cultural relations, ideas and practices as well as physical structures and the environment*" (2017, p. 72).

In situ, such networks produce "hydrosocial territory," a heuristic most clearly described as "*the contested imaginary and socio-environmental materialization of a spatially bound multi-scalar network in which humans, water flows, ecological relationship, hydraulic infrastructure, financial means, legal-administrative arrangements and cultural institutions and practices are interactively defined, aligned and mobilized through epistemological belief systems, political hierarchies and naturalizing discourses*" (Boelens *et al.* 2016, p. 2). Hydrosocial territories have been used in a number of ways throughout the hydrosocial literature. Wilson's (2014) qualitative study of the indigenous Koyukon Athabascan people's lifeways in Ruby, Alaska, draws on the theoretical development of territory in hydrosocial literature to explore the issues of sovereignty and traditional lands in the context of indigenous governance. She found that the Koyukon gained hegemony through both demands for state recognition and by practicing an alternative water monitoring program outside of recognition. Similarly, Peloso and Harris use semi-structured interviews to analyze the potential for a participatory water governance that takes into account "*existing social networks and community governance mechanisms*" in Ashaiman, Ghana (2017, p. 24). They concluded that "*modern water*" is not as effective as bundling water with social welfare to produce participatory governance through different approaches to socio-institutional arrangements. Using survey data pertaining to decision-making around water use in urban Australia, Farrelly and Brown (2014) took a path analysis toward understanding more mixed water infrastructures incorporating water recycling technologies and altering the "hydrosocial contract" to include more civic participation. Their results suggested that diverse infrastructure development could enable "*co-governance, co-design, and co-management*" of complex water systems.

Amid the multi-scalar networks that produce hydrosocial territories, perhaps the most prevalent analytical heuristic in the literature is that of Foucault's "governmentality." Governmentality involves the linkage between mentality and government, the rational underpinnings of that bind society and political sovereignty, and is understood in different ways by hydrosocial theorists (Foucault 2011). Vallardes and Boelens understand governmentality as emerging through four interrelated "*arts of government*" (Foucault 2011, p. 261): the regime of Truth, of sovereign power, of disciplinary power, and of neoliberal power (Valladares and Boelens 2019). Based on these "arts," power comes not just from the top down but also from the bottom up and horizontally, diffuse and productive. In tandem with governmentality, hydrosocial literature also calls on Foucault's notion of biopolitics based, for instance, "*on the categorization, quantification, and knowledge/power formation of urban residents in an attempt to govern their behavior*" (Bakker 2014, p. 283). Hence, the biopolitics of water involve its privatization (Bakker 2014), the removal of water management from people with experiential knowledge (Sarmiento *et al.* 2019), its theft by disenfranchised people (Meehan 2013), its epistemological construction (Hommes *et al.* 2016), its usage as a productive force (Duarte-Abadia *et al.* 2019), its administration through large-scale infrastructure programs (Rogers *et al.* 2016), and the societal rescaling and conflict that emerge around it (Duarte-Abadia and Boelens 2016, Workman 2019).

As a result of the focus on power and scale, we find that hydrosocial studies have traditionally centered on water governance pertaining to general demand, rather than quality, extreme events, and ecosystem services (Fig. 2). The majority of these studies use content analysis of extant theoretical and legal documents to describe how different places fit or do not fit the conceptual models previously established for hydrosocial territories, waterscapes, and other heuristics for understanding power relations across multiple scales in complex, adaptive human–water systems. As well, we identified 15 articles that discuss semi-structured interviews in their abstracts, while seven abstracts disclose use of ethnographic observation. Four articles utilized survey data. Interestingly, 2019 also saw more quantitative methods used than usual, including groundwater modeling (Castilla-Rho *et al.* 2019) and system dynamics (Maxwell *et al.* 2019).

Origins of socio-hydrology

Cited 578 times since its publication as of 20 January 2020, socio-hydrology's inaugural article by Sivapalan *et al.* (2012) enjoined hydrologists to expand their analyses of water systems in a fashion that includes human action as "*part and parcel of water cycle dynamics*" rather than an external force (p. 1271). Importantly, the authors situated their research under Integrated Water Resources Management (IWRM), arguing that "*socio-hydrology is the fundamental science underpinning the practice of IWRM*" (p. 1271). Perhaps most importantly, the observations of socio-hydrology reject stationary models in favor of explorations of "*the co-evolution and self-organization of people in the landscape, also with respect to water availability*" (p. 1271). Hence, socio-hydrology grew to bolster IWRM with a rapidly increasing theoretical corpus that included the fundamental concepts of co-evolution and complexity amid the co-evolution of human and water systems (Fig. 1).

It is important to note that Sivapalan and the other authors of the pioneering socio-hydrology article based their development on prior research, particularly regarding eco-hydrology. Beginning with Frank Geels's (2005) study of the co-evolution of hydrology and Dutch society, the authors call on the sociology of technology and a whole literature of "*socio-technological systems*," which also influenced hydrosocial theorists (Freeman and Perez 1988, Geels 2005). Similarly, socio-hydrology's innovators call on Kallis (2010) "*vicious cycles*" in Athenian water use, which describes "*the hydro-environmental geography of Athens*" over time as economic investments drove structural changes that correlated with social transformations (p. 801). For his part, Kallis identified a number of similar studies dating back to the early 1990s, placing the marker for socio-hydrology's precursive period further back into the archives (and mingling with hydrosocial theory through socio-technological systems research).

In an interesting reaction to the Sivapalan *et al.* (2012)'s inaugural "socio-hydrology" article, Sivakumar argued that "socio-hydrology" was "*not a new science, but a recycled and reworded hydrosociology*" (Sivakumar 2012, p. 3788). Referring to the work of Falkenmark and Wildstrand from the late 1970s, Sivakumar insisted that "*any new study on the dynamics and co-evolution of coupled human–water systems can only be, at best, an addition to the science of hydrosociology rather than a new science by itself*" (p. 3789). Pande and Sivapalan responded indirectly by noting Falkenmark's contributions and calling for their expansion by making "[l]*ong-term socio-economic (such as population, wealth, etc.) and water infrastructure scenarios (e.g. demand projections and water policy)*" endogenous to the study of coupled human–water systems (2017, p. 2). Indeed, the concepts presented in contemporary socio-hydrology deserve discussion on their own merits rather than subsumption within the corpus of hydrosocial studies based on the latter's earlier appearance for the following three reasons:

(1) both hydrosocial theory and socio-hydrology have progressed from Falkenmark's late 1970s work in hydrosociology;
(2) socio-hydrology and contemporary hydrosocial theory are discrete subfields of the study of coupled human–water systems with numerous differences; and
(3) calling socio-hydrology "hydrosociology" would only instill greater confusion over apparent conflation with contemporary hydrosocial theory, which despite the increasing amount of overlap and synthesis, should still be considered on its own merits.

Having rejected the categorical subsumption of socio-hydrology into hydrosociology or hydrosocial theory, we must further investigate socio-hydrology to understand the challenges of ongoing efforts at synthesizing the research surrounding coupled human–water systems.

Reviewing socio-hydrology

While hydrosocial research sometimes includes statistical models, causal feedback diagrams, and contemplation on system dynamics, socio-hydrology tends to incorporate those methods far more regularly [Table 1]. Several review articles of socio-hydrology already exist. Nusser, having first published the term "socio-hydrology" in 2012 without calling for a "new science," produced a brief, approving review of the literature (2015). As well, Xu *et al.* (2018) offer a generous review calling for the inclusion of a social science perspective without necessarily making a direct link to hydrosocial theory or mapping out a spatial and topical analysis of the research. Other reviews of models and theoretical discourses, such as that produced by Di Baldassarre *et al.* (2016), offer incisive meditations on uncertainty and precision in socio-hydrology that are often outward-looking but lack specific focus on the other leading school of thought regarding human–water systems.

By contrast, Massuel *et al.* (2018) call for further interdisciplinary research recognizing that "*the hydrological model itself becomes (only) one element in the socio-hydrological approach rather than its end purpose*" (p. 2518). Indeed, it appears that the further socio-hydrological analysis develops, the more intellectual space becomes available to address phenomena overlapping with hydrosocial theory, such as the *institutional complexity effect*, whereby "*coupled human–water systems often evolve in ways that add more complex infrastructure and governance*

Table 1. Studies utilizing different types of socio-hydrological and hydrosocial models.

Agent based models			System dynamics		
Reference	Study area	Problem	Reference	Study Area	Problem
Abebe *et al.* (2019)	Sint Maarten	Flood risk management	Barendrecht *et al.* (2019)	Dresden	Flood Settlement
Farjad *et al.* (2017)	Alberta, CA	River flow modeling	Borgomeo *et al.* (2018)	Bangladesh	Water Poverty Trap
Klassert *et al.* (2015)	Amman, Jordan	Water consumption	Duran-Encalada *et al.* (2016)	MX-US border	Population and consumption
Walker *et al.* (2015)	Theoretical	Human-water systems	Elshafei *et al.* (2015)	West Australia	Water balance co-evolution
Tesfatsiona *et al.* (2017)	Squaw Creel, Iowa	Climate and decision making	Feng *et al.* (2016)	Hehuang, China	Power generation and water supply
Bakarji *et al.* (2017)	Theoretical	Decision support	Gober and Wheater (2015)	Saskatchewan	Irrigation and water security
Aert *et al.* (2018)	Review	Disaster Risk Reduction	Gunda *et al.* (2018)	New Mexico	Acequia water management
O'Connell and O'Donnell (2014)	Israel/Palestine	Floods, virtual water	Jeong and Adamowski (2016)	South Korea	Irrigation and SWAT model
Wens *et al.* (2019)	General	Drought risk perception	Liu *et al.* (2015)	Tarim River, China	Socio-economics and resources
			Mehta *et al.* (2015)	Bangalore, India	Urban metabolism and Enviro Justice
			Pande and Savenije (2016)	Global	Water metabolism
			Roobavannan *et al.* (2017)	Review	Generalization of system dynamics
			Srinivasan (2015)	Chennai, India	Consumer wells, dynamic infrastructure
			Turner *et al.* (2016)	New Mexico	Acequia water management
			Wheater and Gober (2015)	Saskatchewan	Non-stationary and vulnerability analysis
			Xu *et al.* (2018)	Theoretical	Systemic risk with adaptive governance

arrangements to reduce hydrological variability and increase system performance" (Di Baldassarre *et al.* 2019, p. 6335). Continuing on this theme, Di Baldassarre *et al.* (2019) call for mixed methods approaches that address the challenge of "*different epistemologies, research strategies, and axiologies of qualitative and quantitative approaches*," inclusive of "*central themes of scholars in critical water studies and political ecology such as the role of ideology, and regional landscapes and the uneven distribution of costs and benefits thereof*" (p. 6344). As the International Association of Hydrological Science's current focus on human–water systems vis-à-vis the Panta Rei decade of scientific inquiry, the present review hopes to follow this line in the extant research and to point to novel developments going forward.

According to its inaugural paper, socio-hydrology focuses on three categories: historical socio-hydrology, comparative socio-hydrology, and process socio-hydrology. Excluding review papers and theoretical discourses from our study, we discovered 81 process socio-hydrology articles and 49 comparative socio-hydrology articles, with only 12 historical studies in existence. Authors most central to the field, with the most co-authorships, include Sivapalan, Blöschl, Di Baldassarre, Biglione, and Srinivasan (Xu *et al.* 2018). The focus on process studies illustrates socio-hydrology's greater investment in practical applications of specific case studies using quantitative methods to solve existing problems. Our Web of Science search found that methods in socio-hydrology draw largely from physical and statistical models, especially system dynamics and agent-based models – with the exception of theoretical and review articles and some 14 articles utilizing sociological methods like interviews, focus groups, and surveys (four of those occurring in the last year).

The majority of socio-hydrological studies come from Asia (28), with nearly half of those coming from South Asia. Only five studies could be located in Africa. The most common focus is flood management and risk (68 articles), with other important topics including drought (29), and groundwater and irrigation (35). Fewer articles focused on governance, with socio-hydrology taking a sharper focus on water management than water governance in most years. One might conclude that socio-hydrology's practical approach tends to be more amenable to the study of water and climate extremes than water use, ecosystem services, or water quality because socio-hydrologists often take more of an engineering approach than a humanistic one. However, there is some annual variation in the number of study topics, suggesting that broader governance problems can also benefit from socio-hydrological approaches (Fig. 3). Moreover, although scholars tend to think of hydrosocial studies as more theory-intensive, socio-hydrology literature involves a higher proportion of articles without case studies, including review articles, discussions on different forms of models, and broader studies across different continents.

Despite its impressive focus on quantitative analysis based on complex physical and statistical models, socio-hydrology also incorporates critical analyses of modeling methodology. Srinivasan put the challenges of modeling complex adaptive systems in a manner reminiscent of hydrosocial critique:

As new technologies develop, users adapt to unreliable water supply. Adaptive responses by humans (acting individually and collectively) in turn may alter the watershed hydrology and consequently water availability. These bi-directional feedbacks often result in unexpected emergent behavior. Many water managers fail to account for these complexities (Srivnivasan 2015, p. 786).

Recognizing the unpredictable nature of such feedbacks, Srinivasan identified three approaches: a "toy" model that is generalizable but not specific, a joint agent-hydrologic model that is accurate for specific, predicted conditions, and a "stylized" model that represents essential characteristics of a system without exact calibration and validation (Srivnivasan 2015).

An interesting "stylized" model can be found in Ferdous *et al.* (2018), who attempted to bridge the gap between generic models and specific case studies with what the authors call "*socio-hydrological spaces*." Here, the authors hoped to move a step

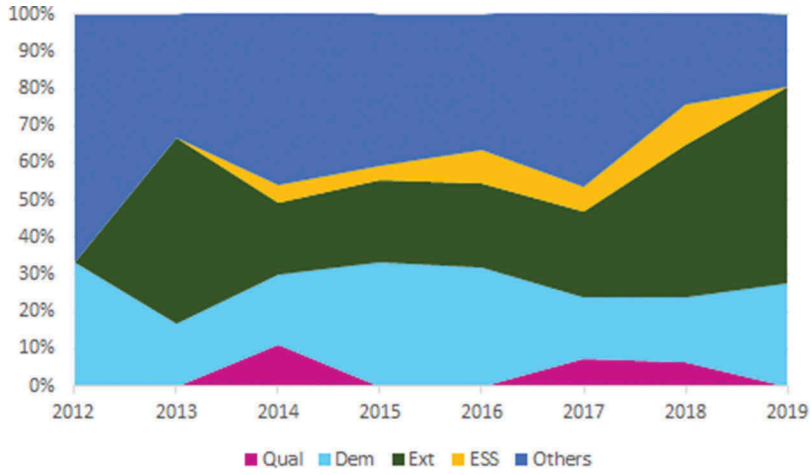

Figure 3. Percentage of socio-hydrology studies by topic: Extremes (purple), Demand/Use (blue), Quality (orange), and Ecosystems Services (green).

beyond conceptual models by utilizing site-specific narratives to articulate the socio-hydrological space, backed by statistical analysis, without providing a formal model. The authors analyzed socio-hydrological spaces as discrete constructions of human responses to physical phenomena like floods – namely the influence of a population's choice to "fight" floods by creating infrastructure or to "adapt" to floods by living with them is correlated with property damage and flood impact during extreme events. Using statistical and narrative analysis, the authors showed that the two different groups choosing to "fight" or "adapt" could be empirically understood as distinct, and those who "adapt" experience less damage. Thus, the authors contest that they can better discern co-evolved human systems that maintain adaptive approaches to hazards from those that are less adaptive. Indeed, lessons from socio-hydrological spaces can also be applied to hydrosocial territories in order to create comparative studies of multiscalar networks in different sites and their capacities to "fight" or "adapt" to different modes of governance and forms of hazards.

Sociological methods are becoming increasingly important but remain less commonly used than physical or statistical modeling in socio-hydrology. Houston et al. engaged a survey of residents in Newport Bay Estuary to examine correlations between "*nonspatial perceptions of dread*" and a spatial understanding of areas prone to flooding (Houston *et al.* 2019, p. 347). In their study of riparian farmers in Tunisia, Ogilvie *et al.* (2019, p. 17) deployed both questionnaires and semi-structured interviews, mixing quantitative and qualitative methods in a "*multi-scalar interdisciplinary approach*" to understanding diversification of agricultural practices and resilience to drought. Similarly, Nüsser *et al.* (2019) used field surveys and interviews, along with remote sensing, to assess the efficacy of artificial glaciers in Ladakh, India, in hydrological terms and according to smallholders' perceptions. It should be noted that the use of these sociologically based methods often correlates to theoretical overlap with hydrosocial theory.

Meanwhile, some argument exists within the socio-hydrological literature over the nature and importance of quantitative modeling methods. Bekchanov *et al.* (2017) argued that Water Economy Models can help drive ecosystem services with the addition of more precise economic data. Rather than critique the usage of models the authors favor them as neutral tools to infer water consumption in such a way as to help steer policy depending on the quality of data. In an important rejoinder to the discussion across sub-fields, Melsen *et al.* (2018, p. 1435) claimed that models are "*uncertain, subjective and a product of the society in which they were shaped*". While their work is largely associated with hydrosocial theory, the authors' intervention in the subfield of socio-hydrology indicates important shared values in the midst of difficult methodological problems.

In their response to Melsen *et al.* (2018), six socio-hydrologists including Sivipalan and Blösch stated, "[w]*hile we reiterate that, despite acknowledged shortcomings, the enterprise of integrating societal feedbacks into hydrological models is beneficial in prediction and adaptive management, we also agree with the sentiments of the authors*" (Srinivasan *et al.* 2018, p. 1444). As recent modeling efforts become even more refined and complex, for instance incorporating machine learning to simulate adaptive irrigation (Mewes and Schumann 2019), efforts to broaden socio-hydrology beyond modeling have also advanced. For instance, Borga *et al.* (2019) worked to implement interdisciplinary collaboration to create post-flood surveys that include eyewitness interviews along with physical data, and Kam *et al.* (2019) used Google Trends to study drought awareness in California from 2011 to 2017. Along with this apparent agreement about broadening methodological approaches, we also found areas of study that appear to integrate hydrosocial theory and socio-hydrology in new ways, for instance with hydroeconomics (Jaeger *et al.* 2017, Müller and Levy 2019). It would appear that the two subfields uncovering coevolutionary dynamics of human–water systems are themselves evolving together through the exchange of ideas and productive critiques.

Bridging the gap

These recent developments elucidate how socio-hydrology, understood as a "positivist" science by Pande and Sivapalan (2017), can also extend to other approaches that also offer

generalizable results. Here, positivism, or (post)positivism, may not preclude the materialist or even post-structuralist approaches of hydrosocial theory. Socio-hydrology can also take biocentric perspectives or an approach closer to current fields of thought like Object Oriented Ontology or Assemblage Theory that identify human action as an important causal agent among others in environmental change (e.g. Delanda 2016). The point of socio-hydrology, then, is not necessarily anthropocentric but an effort to make short- and long-term human behavior endogenous to a hydrologic system in approximately the same way hydrosocial theorists ideate the hydrosocial cycle.

Indeed, socio-hydrologists have mulled over and incorporated the implications of the same foundational theories as hydrosocial studies, such as Actor-Network Theory (ANT). In their call for the inclusion of ANT progenitor Bruno Latour's theoretical insights regarding networks in agriculture archaeology in order to show "*how human agency shapes relationships and institutions*," Ertsen *et al.* (2015, p. 1381) revealed how cross-fertilization between theoretical approaches between socio-hydrology and hydrosocial research has been put in practice. Similarly, Lane drew extensively on Latour and his associate Michel Callon to promote "*[greater] public involvement in scientific practice*" as a way of improving the translation of science into policy in flood risk management (Lane 2014, p. 935). Yet, it is not merely the usage of ANT that aligns hydrosocial and socio-hydrological theory but their shared interest in challenging the limits of even the most up-to-date models in describing complex and unpredictable socio-natural systems to extend empirical research designs with broader, systemic assessments.

Epistemological studies, more clearly at the bedrock of hydrosocial theory, are also increasingly vital to socio-hydrology – a tendency that explicitly involves integration with hydrosocial theory (Di Baldassarre *et al.* 2019, p. 6344). For example, Wescoat (2013) examined the "*duty of water*" – a discursive construct familiar to hydrosocial researchers – through an epistemological approach to the fundamental assumptions made by British irrigation engineers and planners who helped set the standard for US water systems in the west. Similarly, Strickert *et al.* (2015) also called for the usage of Cultural Theory in ideating societal conflict and risk pertaining to drought. Van Loon *et al.* (2016) described human action as an important driver of climate change, calling for definitions of drought to include differentiation between human-modified drought and other drought conditions. Moreover, Callegary *et al.* (2018, p. 60), in identifying "*the utility of social science in applied hydrologic research*," also identified multi-scalar networks without naming them, describing complex relationships between stakeholders, transboundary legal arrangements, and institutional frameworks across both local and global scales. As well, Mukherji *et al.* (2019) provided a recent discussion of epistemological understandings of power, politics, and intersectionality in the cryosphere that would be at home in a hydrosocial special issue. The authors also cited researchers in critical geographies of water like Norman and Bakker, calling for greater focus on "*socio-political and historical information, governance, legal and institutional frameworks, cultural sensitivity, communication and stakeholder engagement among others*" (p. 71). These instances all illustrate tendencies of cross-over that traverse the boundaries dividing subfields, incorporate different, intersecting disciplines, and implement new critical developments. Perhaps the ongoing tendencies toward integration manifest a partial result of a process of critique and learning undertaken between subfields that draws them into a multi-part assemblage focusing on different, complimentary systems with broad and perhaps increasing overlap.

Further evidence of this overlap, or emergent syntheses, between hydrosocial theory and sociohydrology can be found in more recent, innovative approaches. Investigating the power relations within stakeholder relations relative to groundwater contamination, studies in the new subfield of socio-hydrogeology merge socio-economic assessment and hydrogeology, leading to the identification of important variables otherwise ignored (Re 2015, Re *et al.* 2017). Similarly, Di Pelino *et al.* (2019) argue for a coupled human–water approach to public health, which fuses the coupled-systems approach of socio-hydrology, socio-hydrogeology, while "*disaggregating into specific elements and behaviours and avoid over-generalization*" in the hydrosocial fashion, as well (p. 7). This ecohealth approach, the authors explain, would offer a "*more comprehensive understanding of human health within the context of environmental issues*" (p. 8). Socio-hydrology and hydrosocial theory are already engaging in a process of interfacing while maintaining haecceity and producing new and fascinating currents that will continue to contribute to the fecund discussions producing new and interesting subfields beyond the Panta Rei period.

Even the spatial distribution of studies seems to indicate symbiosis. While hydrosocial studies are concentrated in Europe and Latin America (Fig. 4), socio-hydrology is strongest in Asia (Fig. 5) While socio-hydrology incorporates more research from Northern Africa, hydrosocial studies tend to fall in sub-Saharan Africa. Indeed, the maps of the two studies are symmetrical in compelling ways. At the same time, we can see that there is a considerable lack of forays into water quality and ecosystem services, indicating perhaps an avenue for expansion in future collaborative studies.

Although they seem quite different, then, our study found that hydrosocial research and socio-hydrology share many of the same common presuppositions, inclusive of a conscientious understanding of coupled human–water systems based on the impetus to promote egalitarian resource management by integrating multifarious stakeholders into the planning and implementation of policy. Socio-hydrology appears to deploy more quantitative methods, using suites of statistical and physical models to understand potential scenarios, but Srivnivasan's (2015) development of "stylized models" provides an invitation to hydrosocial researchers to mix their methods. As well, hydrosocial approaches will increasingly develop and cross-pollinate with socio-hydrological research across different scales in order to produce still more robust and useful research. Furthermore, hydrosocial theory's reliance on marxian political economy remains malleable given its heterodox form and calls for a multiplicity of epistemological engagements with different systems, not to be seen as a limitation but as a wellspring for future discussion and vigorous debate

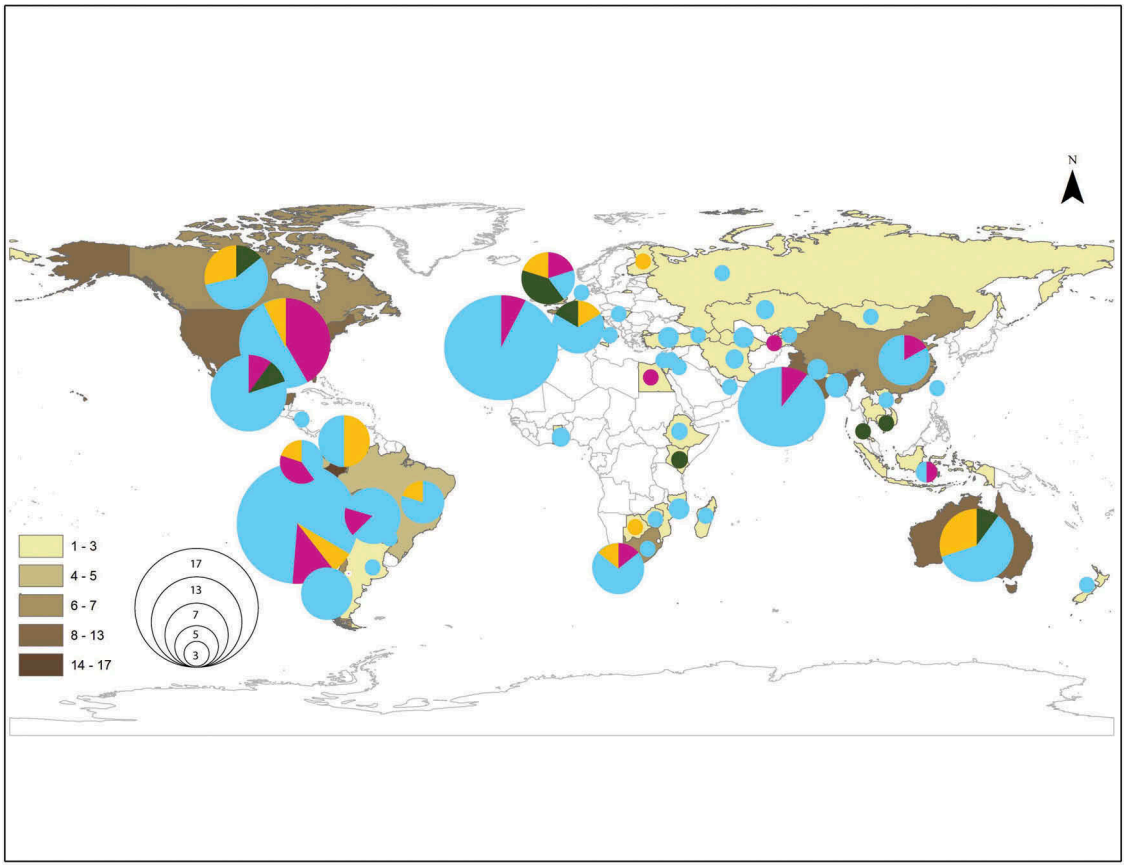

Figure 4. Map showing distribution of hydrosocial studies via January 2020 topic search using Web of Science. Extremes (green), Demand/Use (blue), Quality (purple), Ecosystem Services (yellow).

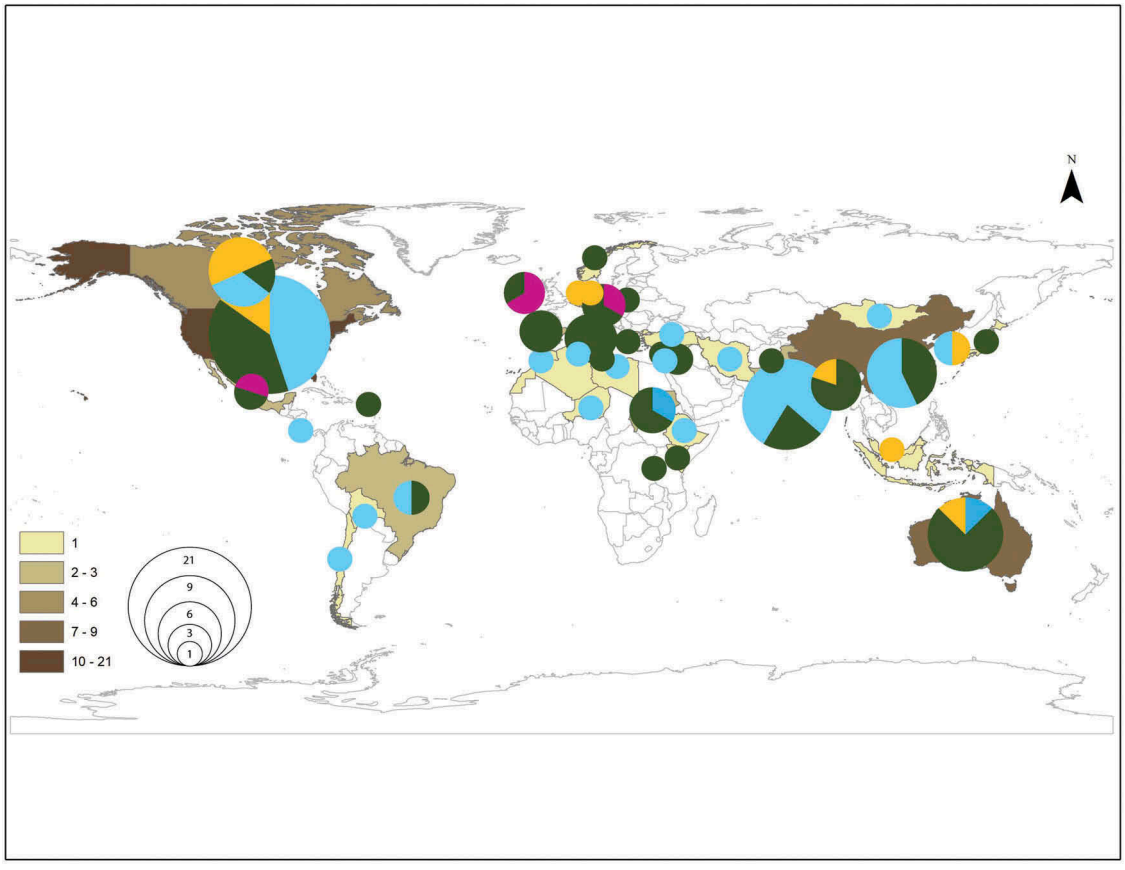

Figure 5. Map showing distribution of socio-hydrology studies via January 2020 topic search using Web of Science. Extremes (green), Demand/Use (blue), Quality (purple), Ecosystem Services (yellow).

among analysts of human–water systems for which historical socio-hydrology will provide a significant source.

Conclusions

Our review of the 419 articles from socio-hydrology and hydrosocial or hydro-social studies suggests that the two subfields have grown closer together and are adapting to critiques, although the publication outlets remain different. Hydrosocial studies increasingly emerge from a diverse array of sites, gathering localized observations through rigorous methods, while socio-hydrology's openness to assessing the implicit biases in hydrological modeling and navigating theoretical nuance invites active cross-fertilization of methods and theories. As well, a number of overlapping positions can clearly be seen:

(1) The conceptual exploration of socio-hydrological spaces, hydrosocial territories, and waterscapes can offer possible transdisciplinary syntheses of practical solutions and theoretical outlooks regarding perceptions of risk, resilience, and adaptivity.
(2) The joint recognition that humans must be considered endogenous to an understanding of a hydrological system that we impact moves in tandem with systems analyses that appreciate feedbacks and "metabolic" processes, opening opportunities for studies that incorporate quantitative and qualitative methods.
(3) Situated epistemologies produce conditions of power and scale that impact hydrosocial/socio-hydrological systems, while the collaboration of hydrosocial and socio-hydrological studies can help develop conscientious solutions to complex problems in both the short and long term.

Socio-hydrology and hydrosocial theory stem from different ways of reconciling old development regimes to transformative, emergent scientific approaches. While hydrosocial theory has evolved from fields like urban political ecology, it is by no means restricted to those roots, employing a more rhizomal distribution across disciplines. Similarly, socio-hydrologic research looks to myriad places to test different strategies for managing water in ways that align with complex, adaptive, and dynamic ecosystems amid climate change in the Anthropocene. Indeed, we point not only to collaboration across disciplines but to deeper integration between them through the development of more holistic studies that might be seen as an overlapping set in a Venn Diagram. Such studies model scenarios involving the implications of power relations for coupled human–water systems across multiple scales to identify practical solutions to extant problems without losing sight of their boundary conditions (Evers *et al.* 2017). Thus, we do not seek to promote subsumption but rather applaud the ongoing assemblages of methods and epistemological approaches that harmonize the different subfields into compelling and robust studies without losing their integral distinctions in keeping with what Rusca and Di Baldassarre call "*interdisciplinary resource geography*" (2019, p. 8–10)

Further research might extend the contemporary trends into a more integrated approach, using subfields of hydrosocial studies and socio-hydrology through the oft-neglected lens of exurban political ecology, which will help provide some insight into the population trends and the responses of watershed ecosystems on which they depend to climate change (McCarthy 2002, McKinnon and Hiner 2016, McKinnon *et al.* 2017). Here, the "stylized" model may offer the greatest opportunity for hydrosocial scientists to use their understanding of power relations toward producing sound, practical proposals for hydrosocial territories while engaging with the dynamism of complex, adaptive systems (Milly *et al.* 2008).

In turn, by making research more sociologically oriented, with rigorous applications of qualitative methods, socio-hydrologists can make their interests more widespread. Furthermore, by branching out and gaining insight from within anthrospheres, socio-hydrologists will have a better opportunity to share and enrichen their ideas, particularly regarding IRWM. Socio-hydrologists may be wise to publish more in social science journals and solicit water resources management articles from social scientists interested in pressing water issues. Finally, greater collaboration with social scientists, which offers transformative frameworks for modeling and translating research material to various stakeholders, would help bridge the translation gap between science and policy (Xu *et al.* 2018). Indeed, the progress in both socio-hydrology and hydrosociology is encouraging. As climate change produces new and difficult problems, adaptive research communities have drawn these distinct subfields closer together, resulting in a vital cross-pollination that contains transformative potential for future scholars and practitioners to explore together (Ajibade and Adams 2019).

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